

1. Determine the convergence of each of the following series.

(a) $\sum_{k=0}^{\infty} \left(\frac{e}{\pi}\right)^k$ is convergent; it's a geometric series with $r = \frac{e}{\pi} < 1$

(b) $\sum_{k=1}^{\infty} \frac{2}{k\sqrt{k}}$ is convergent; it's a p -series with $p = 1.5 > 1$

(c) $\sum_{k=1}^{\infty} \frac{1}{k2^k}$ is convergent by ratio test

$$\lim_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} = \lim_{k \rightarrow \infty} \frac{k 2^k}{(k+1)2^{k+1}} = \frac{1}{2} \lim_{k \rightarrow \infty} \frac{k}{k+1} = \frac{1}{2} < 1$$

(d) $\sum_{k=2}^{\infty} \frac{1}{\sqrt{k}-1}$ is divergent by comparison test

$$\sqrt{k}-1 < \sqrt{k} \Rightarrow \frac{1}{\sqrt{k}-1} > \frac{1}{\sqrt{k}} \text{ and } \sum_{k=2}^{\infty} \frac{1}{\sqrt{k}} \text{ is a divergent } p\text{-series}$$

(e) $\sum_{k=2}^{\infty} \frac{(k-1)!}{(k+1)^2}$ is divergent by ratio test

$$\lim_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} = \lim_{k \rightarrow \infty} \frac{k! (k+1)^2}{(k+2)^2 (k-1)!} = \lim_{k \rightarrow \infty} k \left(\frac{k+1}{k+2}\right)^2 = \infty$$

(f) $\sum_{k=2}^{\infty} \frac{2^{k+2}}{\ln k}$ is divergent by ratio test

$$\lim_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} = \lim_{k \rightarrow \infty} \frac{2^{k+3} \ln k}{\ln(k+1) 2^{k+2}} = 2 \lim_{k \rightarrow \infty} \frac{\ln k}{\ln(k+1)}$$

$$= 2 \lim_{k \rightarrow \infty} \frac{\frac{1}{k}}{\frac{1}{k+1}} = 2 \lim_{k \rightarrow \infty} \frac{k+1}{k} = 2 > 1$$

2. Let $|r| < 1$. By considering the partial sums S_n and S_{n-1} , show that $\sum_{k=0}^{\infty} r^k = \frac{1}{1-r}$.

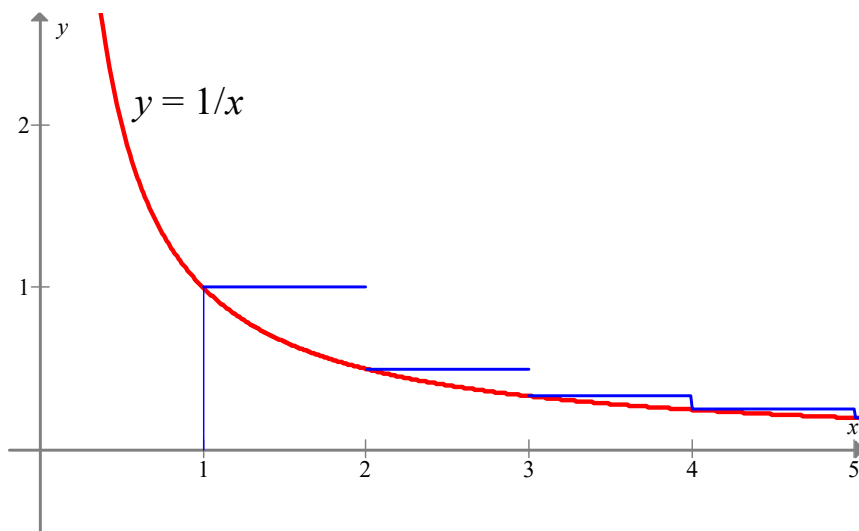
$$\text{Let } S = \lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} S_{n-1} = \sum_{k=0}^{\infty} r^k$$

$$S_n = 1 + r + r^2 + \dots + r^{n-1} + r^n$$

$$S_{n-1} = 1 + r + r^2 + \dots + r^{n-1}$$

$$S_n - r S_{n-1} = 1 \Rightarrow S - rS = 1 \Rightarrow (1-r)S = 1 \Rightarrow S = \frac{1}{1-r}$$

3. Show that $\sum_{k=1}^{\infty} \frac{1}{k}$ is divergent by comparing it with the improper integral $\int_1^{\infty} \frac{1}{x} dx$.



Referring to the graph above, the area of the shaded region consisting of n many

rectangulars is the partial sum $S_n = \sum_{k=1}^n \frac{1}{k}$, which is larger than the area under the curve

of $y = \frac{1}{x}$ over the interval $[1, n+1]$. Hence,

$$S_n = \sum_{k=1}^n \frac{1}{k} > \int_1^{n+1} \frac{1}{x} dx = [\ln x]_1^{n+1} = \ln(n+1)$$

and this implies

$$S = \lim_{n \rightarrow \infty} S_n \geq \lim_{n \rightarrow \infty} \ln(n+1) = \infty,$$

which means that the harmonic series is divergent.

4. Find the interval of convergence of each of the following power series.

$$(a) \sum_{k=1}^{\infty} \frac{(x-1)^k}{\sqrt{k}}$$

$$\lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right| = \lim_{k \rightarrow \infty} \left| \frac{(x-1)^{k+1} \sqrt{k}}{\sqrt{k+1} (x-1)^k} \right| = |x-1| \lim_{k \rightarrow \infty} \left(\frac{k}{k+1} \right)^{1/2} = |x-1|$$

$$\lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right| < 1 \Rightarrow |x-1| < 1 \Rightarrow -1 < x-1 < 1 \Rightarrow 0 < x < 2$$

$$x = 0 \Rightarrow \sum_{k=1}^{\infty} \frac{(0-1)^k}{\sqrt{k}} = \sum_{k=1}^{\infty} \frac{(-1)^k}{k^{1/2}} \text{ is a convergent alternating } p\text{-series}$$

$$x = 2 \Rightarrow \sum_{k=1}^{\infty} \frac{(2-1)^k}{\sqrt{k}} = \sum_{k=1}^{\infty} \frac{1}{k^{1/2}} \text{ is a divergent } p\text{-series}$$

Thus, the interval of convergence is $0 \leq x < 2$

$$(b) \sum_{k=1}^{\infty} \frac{2^k x^{2k}}{k}$$

$$\lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right| = \lim_{k \rightarrow \infty} \left| \frac{2^{k+1} x^{2(k+1)} k}{(k+1) 2^k x^{2k}} \right| = |2x^2| \lim_{k \rightarrow \infty} \frac{k}{k+1} = 2x^2$$

$$\lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right| < 1 \Rightarrow 2x^2 < 1 \Rightarrow x^2 < \frac{1}{2} \Rightarrow -\frac{1}{\sqrt{2}} < x < \frac{1}{\sqrt{2}}$$

$$x = -\frac{1}{\sqrt{2}} \Rightarrow \sum_{k=1}^{\infty} \frac{2^k \left(-\frac{1}{\sqrt{2}}\right)^{2k}}{k} = \sum_{k=1}^{\infty} \frac{2^k \left(\frac{1}{2}\right)^k}{k} = \sum_{k=1}^{\infty} \frac{1}{k} \text{ is a divergent harmonic series}$$

$$x = \frac{1}{\sqrt{2}} \Rightarrow \sum_{k=1}^{\infty} \frac{2^k \left(\frac{1}{\sqrt{2}}\right)^{2k}}{k} = \sum_{k=1}^{\infty} \frac{1}{k} \text{ is a divergent harmonic series}$$

Thus, the interval of convergence is $-\frac{1}{\sqrt{2}} < x < \frac{1}{\sqrt{2}}$

$$(c) \sum_{k=1}^{\infty} \frac{(2x-1)^{2k+1}}{k^{3/2}}$$

$$\lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right| = \lim_{k \rightarrow \infty} \left| \frac{(2x-1)^{2(k+1)+1} k^{3/2}}{(k+1)^{3/2} (2x-1)^{2k+1}} \right| = |2x-1|^2 \lim_{k \rightarrow \infty} \left(\frac{k}{k+1} \right)^{3/2} = (2x-1)^2$$

$$\lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right| < 1 \Rightarrow (2x-1)^2 < 1 \Rightarrow -1 < 2x-1 < 1 \Rightarrow 0 < 2x < 2 \Rightarrow 0 < x < 1$$

$$x = 0 \Rightarrow \sum_{k=1}^{\infty} \frac{(0-1)^{2k+1}}{k^{3/2}} = \sum_{k=1}^{\infty} \frac{-1}{k^{3/2}} \text{ is a convergent } p\text{-series}$$

$$x = 1 \Rightarrow \sum_{k=1}^{\infty} \frac{(2-1)^{2k+1}}{k^{3/2}} = \sum_{k=1}^{\infty} \frac{1}{k^{3/2}} \text{ is a convergent } p\text{-series}$$

Thus, the interval of convergence is $0 \leq x \leq 1$

$$(d) \sum_{k=1}^{\infty} \frac{x^k}{5^k \sqrt{k}}$$

$$\lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right| = \lim_{k \rightarrow \infty} \left| \frac{x^{k+1} 5^k \sqrt{k}}{5^{k+1} \sqrt{k+1} x^k} \right| = \left| \frac{x}{5} \right| \lim_{k \rightarrow \infty} \left(\frac{k}{k+1} \right)^{1/2} = \left| \frac{x}{5} \right|$$

$$\lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right| < 1 \Rightarrow \left| \frac{x}{5} \right| < 1 \Rightarrow -1 < \frac{x}{5} < 1 \Rightarrow -5 < x < 5$$

$$x = -5 \Rightarrow \sum_{k=1}^{\infty} \frac{(-5)^k}{5^k \sqrt{k}} = \sum_{k=1}^{\infty} \frac{(-1)^k}{k^{1/2}} \text{ is a convergent alternating } p\text{-series}$$

$$x = 5 \Rightarrow \sum_{k=1}^{\infty} \frac{5^k}{5^k \sqrt{k}} = \sum_{k=1}^{\infty} \frac{1}{k^{1/2}} \text{ is a divergent } p\text{-series}$$

Thus, the interval of convergence is $-5 \leq x < 5$

$$(e) \sum_{k=0}^{\infty} \frac{(x-1)^k}{5^k}$$

$$\lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right| = \lim_{k \rightarrow \infty} \left| \frac{(x-1)^{k+1} 5^k}{5^{k+1} (x-1)^k} \right| = \left| \frac{x-1}{5} \right|$$

$$\lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right| < 1 \Rightarrow \left| \frac{x-1}{5} \right| < 1 \Rightarrow -1 < \frac{x-1}{5} < 1 \Rightarrow -5 < x-1 < 5 \Rightarrow -4 < x < 6$$

$$x = -4 \Rightarrow \sum_{k=1}^{\infty} \frac{(-4-1)^k}{5^k} = \sum_{k=1}^{\infty} \frac{(-5)^k}{5^k} = \sum_{k=1}^{\infty} (-1)^k \text{ is a divergent series}$$

$$x = 6 \Rightarrow \sum_{k=1}^{\infty} \frac{(6-1)^k}{5^k} = \sum_{k=1}^{\infty} \frac{5^k}{5^k} = \sum_{k=1}^{\infty} 1 \text{ is a divergent series}$$

Thus, the interval of convergence is $-4 < x < 6$

$$\begin{aligned}
 \text{(f)} \quad & \sum_{k=0}^{\infty} \frac{(x+1)^{2k}}{4^k} \\
 & \lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right| = \lim_{k \rightarrow \infty} \left| \frac{(x+1)^{2(k+1)} 4^k}{4^{k+1} (x+1)^{2k}} \right| = \left| \frac{(x+1)^2}{4} \right| \\
 & \lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right| < 1 \Rightarrow \frac{(x+1)^2}{4} < 1 \Rightarrow (x+1)^2 < 4 \Rightarrow -2 < x+1 < 2 \Rightarrow -3 < x < 1 \\
 & x = -3 \Rightarrow \sum_{k=1}^{\infty} \frac{(-3+1)^{2k}}{4^k} = \sum_{k=1}^{\infty} \frac{(-2)^{2k}}{4^k} = \sum_{k=1}^{\infty} 1 \text{ is a divergent series} \\
 & x = 1 \Rightarrow \sum_{k=1}^{\infty} \frac{(1+1)^{2k}}{4^k} = \sum_{k=1}^{\infty} \frac{2^{2k}}{4^k} = \sum_{k=1}^{\infty} 1 \text{ is a divergent series}
 \end{aligned}$$

Thus, the interval of convergence is $-3 < x < 1$

$$\begin{aligned}
 \text{(g)} \quad & \sum_{k=2}^{\infty} \frac{(-2)^k x^k}{\sqrt{k-1}} \\
 & \lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right| = \lim_{k \rightarrow \infty} \left| \frac{(-2)^{k+1} x^{k+1} \sqrt{k-1}}{\sqrt{k} (-2)^k x^k} \right| = |-2x| \lim_{k \rightarrow \infty} \left(\frac{k-1}{k} \right)^{1/2} = 2|x| \\
 & \lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right| < 1 \Rightarrow 2|x| < 1 \Rightarrow -\frac{1}{2} < x < \frac{1}{2} \\
 & x = -\frac{1}{2} \Rightarrow \sum_{k=2}^{\infty} \frac{(-2)^k \left(-\frac{1}{2}\right)^k}{\sqrt{k-1}} = \sum_{k=2}^{\infty} \frac{1}{\sqrt{k-1}} = \sum_{k=1}^{\infty} \frac{1}{\sqrt{k}} \text{ is a divergent } p\text{-series} \\
 & x = \frac{1}{2} \Rightarrow \sum_{k=2}^{\infty} \frac{(-2)^k \left(\frac{1}{2}\right)^k}{\sqrt{k-1}} = \sum_{k=2}^{\infty} \frac{(-1)^k}{\sqrt{k-1}} \text{ is a convergent alternating } p\text{-series}
 \end{aligned}$$

Thus, the interval of convergence is $-0.5 < x \leq 0.5$

$$\begin{aligned}
 \text{(h)} \quad & \sum_{k=1}^{\infty} \frac{(-1)^k (x-1)^k}{k^2} \\
 & \lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right| = \lim_{k \rightarrow \infty} \left| \frac{(x-1)^{k+1} k^2}{(k+1)^2 (x-1)^k} \right| = |x-1| \lim_{k \rightarrow \infty} \left(\frac{k}{k+1} \right)^2 = |x-1| \\
 & \lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right| < 1 \Rightarrow |x-1| < 1 \Rightarrow -1 < x-1 < 1 \Rightarrow 0 < x < 2 \\
 & x = 0 \Rightarrow \sum_{k=1}^{\infty} \frac{(-1)^k (-1)^k}{k^2} = \sum_{k=1}^{\infty} \frac{1}{k^2} \text{ is a convergent } p\text{-series} \\
 & x = 2 \Rightarrow \sum_{k=1}^{\infty} \frac{(-1)^k}{k^2} \text{ is a convergent alternating } p\text{-series}
 \end{aligned}$$

Thus, the interval of convergence is $0 \leq x \leq 2$

5. Referring to the functions given in Example 2.4.2,
 (a) Use Theorem 2.4.1 to derive the Taylor/Maclaurin series formula for each function.
 (b) Determine the interval of convergence for each power series.

$$(i) \quad f(x) = \frac{1}{1+x} = (1+x)^{-1} \Rightarrow f^{(k)}(x) = (-1)^k k! (1+x)^{-k-1}$$

$$f^{(k)}(0) = (-1)^k k! \Rightarrow f(x) = \sum_{k=0}^{\infty} \frac{(-1)^k k! x^k}{k!} = \sum_{k=0}^{\infty} (-1)^k x^k$$

$$\lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right| = \lim_{k \rightarrow \infty} \left| \frac{x^{k+1}}{x^k} \right| = |x|$$

$$\lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right| < 1 \Rightarrow |x| < 1 \Rightarrow -1 < x < 1$$

$$x = -1 \Rightarrow \sum_{k=0}^{\infty} (-1)^k (-1)^k = \sum_{k=0}^{\infty} 1 \text{ is a divergent series}$$

$$x = 1 \Rightarrow \sum_{k=0}^{\infty} (-1)^k \text{ is a divergent series}$$

Thus, the interval of convergence is $-1 < x < 1$

$$(ii) \quad f(x) = e^x \Rightarrow f^{(k)}(x) = e^x \Rightarrow f^{(k)}(0) = e^0 = 1 \Rightarrow f(x) = \sum_{k=0}^{\infty} \frac{x^k}{k!}$$

$$\lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right| = \lim_{k \rightarrow \infty} \left| \frac{x^{k+1} k!}{(k+1)! x^k} \right| = |x| \lim_{k \rightarrow \infty} \left| \frac{1}{k+1} \right| = 0$$

Thus, the interval of convergence is $-\infty < x < \infty$

$$(iii) \quad f(x) = \ln x \Rightarrow f(1) = 0$$

$$f'(x) = \frac{1}{x} = x^{-1} \Rightarrow f^{(k)}(x) = (-1)^{k-1} (k-1)! x^{-k} \Rightarrow f^{(k)}(1) = (-1)^{k-1} (k-1)!$$

$$f(x) = \sum_{k=1}^{\infty} \frac{(-1)^{k-1} (k-1)!}{k!} (x-1)^k = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} (x-1)^k$$

$$\lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right| = \lim_{k \rightarrow \infty} \left| \frac{(x-1)^{k+1} k}{(k+1) (x-1)^k} \right| = |x-1| \lim_{k \rightarrow \infty} \left| \frac{k}{k+1} \right| = |x-1|$$

$$\lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right| < 1 \Rightarrow |x-1| < 1 \Rightarrow -1 < x-1 < 1 \Rightarrow 0 < x < 2$$

$$x = 0 \Rightarrow \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} (-1)^k = \sum_{k=1}^{\infty} \frac{-1}{k} \text{ is a divergent harmonic series}$$

$$x = 2 \Rightarrow \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} \text{ is a convergent alternating series}$$

Thus, the interval of convergence is $0 < x \leq 2$

(iv) $f(x) = \sin x$

$$f^{(2k)}(x) = (-1)^k \sin x \text{ and } f^{((2k+1))}(x) = (-1)^k \cos x$$

$$f^{(2k)}(0) = 0 \text{ and } f^{((2k+1))}(0) = (-1)^k \text{ for } k = 0, 1, 2, \dots$$

$$f(x) = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!} = \sum_{k=1}^{\infty} \frac{(-1)^{k+1} x^{2k-1}}{(2k-1)!}$$

$$\lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right| = \lim_{k \rightarrow \infty} \left| \frac{x^{2(k+1)-1} (2k-1)!}{(2(k+1)-1)! x^{2k-1}} \right| = |x^2| \lim_{k \rightarrow \infty} \left| \frac{1}{(2k+1)(2k)} \right| = 0$$

Thus, the interval of convergence is $-\infty < x < \infty$

(v) $f(x) = \cos x$

$$f^{(2k)}(x) = (-1)^k \cos x \text{ and } f^{((2k+1))}(x) = (-1)^{k+1} \sin x$$

$$f^{(2k)}(0) = (-1)^k \text{ and } f^{((2k+1))}(0) = 0 \text{ for } k = 0, 1, 2, \dots$$

$$f(x) = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!}$$

$$\lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right| = \lim_{k \rightarrow \infty} \left| \frac{x^{2(k+1)} (2k)!}{(2(k+1))! x^{2k}} \right| = |x^2| \lim_{k \rightarrow \infty} \left| \frac{1}{(2k+2)(2k+1)} \right| = 0$$

Thus, the interval of convergence is $-\infty < x < \infty$

6. Use term-by-term differentiation/integration and appropriate Taylor/Maclaurin series to find the power series representation for $f(x) = \tan^{-1} x$.

$$f'(x) = \frac{d}{dx} \tan^{-1} x = \frac{1}{1+x^2} = \sum_{k=0}^{\infty} (-1)^k x^{2k} = 1 - x^2 + x^4 - x^6 + \dots \text{ for } -1 < x < 1$$

$$f(x) = \int \frac{1}{x^2+1} dx = \int (1 - x^2 + x^4 - x^6 + \dots) dx = C + x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots$$

$$f(0) = C \Rightarrow \tan^{-1}(0) = C \Rightarrow C = 0$$

$$\therefore f(x) = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{2k+1} \text{ for } -1 < x < 1$$